Implementation of the Dehbi (2008) CRW–Langevin Particle Dispersion Model as an ANSYS Fluent UDF: 90° Pipe-Bend Case, Results and Code Review

Technical report, revision 2

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November 2026

Revision note

Revision 1 of this report (March 2026) is kept unchanged in `original/report-dehbi-v1-2026-03.pdf`. Revision 2 uses the same simulations, settings and results; no simulation was re-run. It changes the report as follows:

1. The model equations are written out as they are implemented in `udf/crw_dehbi2008_rev10.c`, with source line references. Revision 1 described the time integration as forward Euler; the code uses a semi-implicit scheme (Section 5.2).
2. The deposition fractions are tabulated from the Fluent DPM summaries that were already reproduced in Appendix B of revision 1.
3. Three statements of revision 1 are withdrawn because the evidence does not support them:
   - that the results show “excellent agreement” with the benchmark (no independent comparison was made, Section 6.3);
   - that the refined mesh achieved “true mesh independence” (no grid study was recorded, Section 3);
   - that the drift term is driven by Reynolds-stress gradients computed by Fluent and that mesh refinement improved results by resolving those gradients. The UDF does not use Reynolds-stress gradients: the wall-normal drift is the analytical derivative of a fitted rms profile, evaluated with u_τ and y^+ of the wall-adjacent cell (Section 5.3).
4. Revision 1 stated that the case replicates the experiment of Pui et al. (1987). It does not (Section 2).
5. A code review of the UDF is added (Section 7). It identifies three implementation issues that affect the physics. They are documented, not corrected, because the reported results were produced with the code as it stands.

1. Objective

The study aims to reproduce the 90° pipe-bend deposition case of Dehbi (2008), who coupled a RANS flow solution with a continuous random walk (CRW) model based on a normalised Langevin equation. Discrete random walk (DRW) models sample velocity fluctuations from locally isotropic turbulence. In inhomogeneous near-wall turbulence this produces a spurious drift of low-inertia particles towards the wall, a violation of the well-mixed condition, so deposition of small particles is over-predicted. Dehbi’s model uses anisotropic near-wall velocity statistics and a drift term proportional to the wall-normal gradient of the rms velocity to restore the well-mixed condition.

The objectives were:

1. Compute the turbulent flow in the bend with the Reynolds stress model (RSM).
2. Implement the CRW–Langevin model as a compiled ANSYS Fluent UDF that replaces the built-in DRW model.
3. Compute the deposition fraction for particle diameters of 5 to 50 µm and compare it with Dehbi (2008) and the experiments of Pui et al. (1987).

2. Geometry and flow conditions

Table 1. Geometry and flow conditions.

Quantity	Value	Source
Pipe diameter D	0.02 m	revision 1, Sec. 2
Bend curvature ratio R/r	4 (R = 0.04 m)	revision 1, Sec. 2
Straight inlet / outlet sections	5D (0.10 m) each	revision 1, Sec. 2
Fluid	air, ρ = 1.225 kg m^{-3}, μ = 1.81 × 10^{-5} Pa s	revision 1, Sec. 4.1
Inlet	uniform velocity U = 3.0 m s^{-1}; inlet turbulence not recorded	revision 1, Sec. 4.1
Reynolds number $Re = \rho UD/\mu$	4.06 × 10^3 (revision 1 states 4080)	computed
Friction Reynolds number $Re_\tau = u_\tau r/\nu$	≈ 140, with u_τ ≈ 0.21 m s^{-1} from the Blasius friction factor	estimated
Outlet	pressure outlet, 0 Pa gauge	revision 1, Sec. 4.1
Walls	stationary, no slip	revision 1, Sec. 4.1



Figure 1. Computational domain: 90° bend with straight inlet and outlet sections.

Difference from the benchmark experiment. Revision 1 stated that the domain "strictly replicates" the experiment of Pui et al. (1987). The turbulent case of that experiment is reported as a tube of diameter 8.51 mm, curvature ratio 5.6 and Re = 10,000 (as summarised by Sakib et al., 2023). The present case (D = 20 mm, R/r = 4, $Re \approx 4.1 \times 10^3$) differs in all three. In addition, a 5D inlet section with a uniform inlet velocity does not produce fully developed pipe flow at the bend. Deposition in bends correlates mainly with the Stokes number, so a comparison with the benchmark is meaningful only at matched Stk, and the effect of the different Re and curvature ratio must be discussed. The flow conditions of the Dehbi (2008) bend case were not available to the author for this revision and must be checked against the paper.

Low-Reynolds-number consequence. At $Re_\tau \approx 140$, y^+ = 100 lies about 7 mm from the wall, 70 % of the pipe radius. The near-wall branch of the model (Section 5.3) is therefore intended to cover most of the cross-section, and its rms profiles, fitted to channel-flow DNS, are applied close to the limit of their range.

3. Mesh

The domain was meshed with structured hexahedral cells. Revision 1 gives a first cell height of 70 µm, a wall $y^+ \approx 0.95$ (Figure 3), 30 inflation layers with a growth ratio of 1.1, and about 5.3×10^6 cells; a baseline mesh of about 2.5×10^6 cells was used first.

Inconsistency to resolve. Thirty layers starting at 70 µm with a growth ratio of 1.1 have a total thickness of $70\,\mu\text{m} \times (1.1^{30} - 1)/0.1 \approx 11.5$ mm, larger than the pipe radius of 10 mm. At least one of these three values must differ from the mesh actually used and will be corrected from the mesh file. It is also not recorded whether $y^+ \approx 0.95$ refers to the cell centroid, as in Fluent's "Wall Yplus", or to the first cell height.

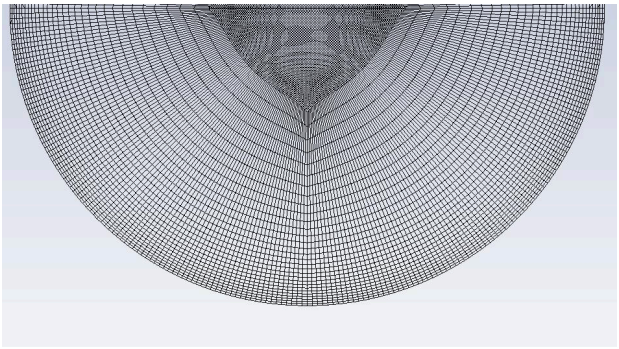
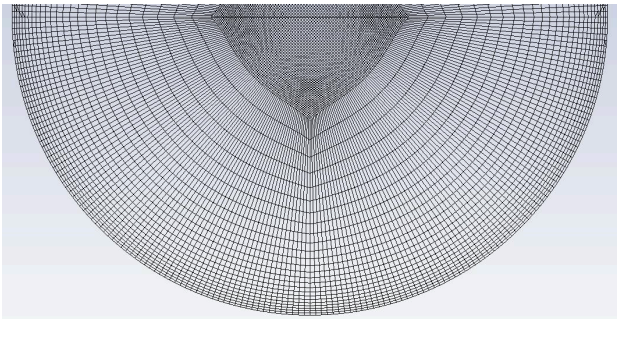
Refined mesh, 5.3×10^6 cells	**Baseline mesh, 2.5×10^6 cells**
	

Figure 2. Cross-section of the bend. The refined mesh adds resolution in the bend and close to the wall.

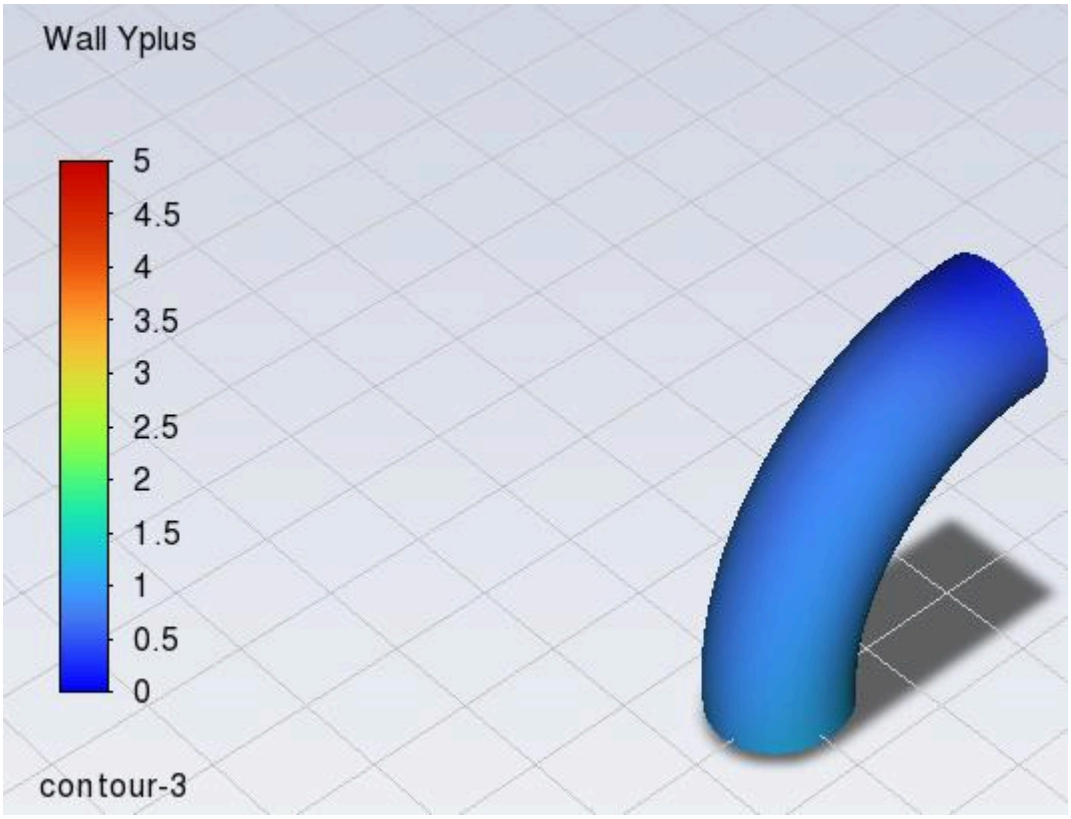


Figure 3. Fluent "Wall Yplus" on the bend surface (colour scale 0 to 5).

Mesh quality histograms (orthogonal quality and skewness) are in `figures/mesh-orthogonal-quality.png` and `figures/mesh-skewness.png`.

Mesh sensitivity. Revision 1 reports that the baseline mesh gave deposition fractions for 5 and 10 µm particles that deviated from the benchmark, while larger particles were insensitive, and that the refined mesh removed the

deviation. The baseline-mesh deposition fractions were not recorded. The study therefore shows that small-particle deposition is sensitive to the near-wall mesh but does not quantify grid convergence. Because of the issue in Section 7.1, part of this sensitivity may come from the thickness of the single cell layer in which the near-wall model is active.

4. Solver settings

Table 2. Solver and discrete-phase settings.

Item	Setting
Solver	ANSYS Fluent 2024 R2, pressure based, steady
Turbulence model	Reynolds stress model, linear pressure-strain, enhanced wall treatment (Figure S1)
Pressure–velocity coupling	SIMPLE
Spatial discretisation	least-squares gradients; second-order pressure; second-order upwind for momentum, k, ε and Reynolds stresses (Figure S2)
Discrete phase	inert spheres, one-way coupling, steady tracking
Injection	surface injection at the inlet, 10,000 particles per diameter, randomised starting points (Figure S5)
Particle density	1000 kg m^{-3} according to revision 1, Sec. 4.3 (to be confirmed, see below)
DPM tracking	accuracy control (tolerance 10^{-5}, 20 refinements), automated Runge–Kutta / implicit scheme (Figure S4)
Built-in turbulent dispersion	disabled; dispersion supplied by the UDF body force
UDF hooks	body force `crw_langevin_force`, time step `crw_adaptive_dt`, 8 particle user scalars (Figure S3); wall boundary condition `dehbi_wall_bc`; on-demand `init_wall_udm`
Deposition	a particle is deposited when `dehbi_wall_bc` stops it at the wall; Fluent reports such particles as “aborted”

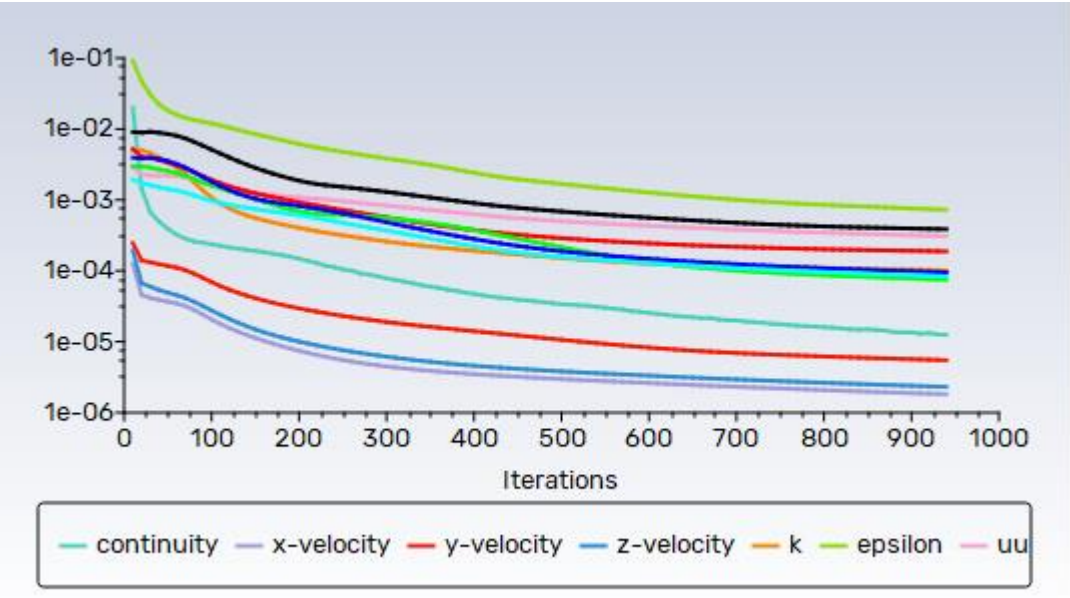


Figure 4. Scaled residuals of the continuous phase.

The scaled residuals decreased to between about 10^{-3} (ε) and 2×10^{-6} over about 950 iterations and were still slowly decreasing. No integral quantity, such as the pressure drop or the wall shear stress, was monitored.

The Fluent dialogs reproduced in revision 1 are kept in `figures/setup/`: S1 viscous model (`viscous-model.png`), S2 solution methods (`solution-methods.png`), S3 DPM UDF hooks (`dpm-udf-hooks.png`), S4 DPM numerics (`dpm-numerics.png`), S5 injection (`injection.png`) and S6 UDF compilation environment (`udf-compilation.png`).

Settings to confirm from the case file.

1. The injection dialog (Figure S5) shows an injection named `particles-10micron` with a diameter of 5×10^{-6} m.
2. The particle material in Figure S5 is `anthracite`, whose Fluent default density is 1550 kg m^{-3}, while revision 1 states 1000 kg m^{-3}. If the default was kept, every Stokes number in Table 3 is 1.55 times larger.
3. The injection velocity in Figure S5 is 0 m s^{-1}, while revision 1 states that particles are released at the gas velocity. With $\tau_p \le 7.7$ ms and a 0.10 m inlet section particles adjust to the gas velocity before the bend, so the effect is expected to be small.
4. With $\Delta t \le 10^{-6}$ s and a residence time of about 0.09 s, each particle needs about 9×10^4 time steps. The DPM "maximum number of steps" setting must exceed this, otherwise particles are stopped early and also counted as "aborted".
5. Gravity and the orientation of the bend are not recorded. The settling velocity $g\tau_p$ is about 7.5 cm s^{-1} at 50 µm, which over the residence time of about 0.09 s corresponds to a displacement of several millimetres in a pipe of 10 mm radius, so gravity affects the larger particles if it was enabled.
6. Three wall zones are present (`bend_90deg`, `wall1`, `wall2`; Figure S5). The deposition fraction in Table 3 counts particles stopped on any of them, including the straight sections, whereas the benchmark experiment reports deposition in the bend. Deposition per zone was not recorded.

5. CRW–Langevin model as implemented

Symbols: u_τ friction velocity, ν kinematic viscosity, $y^+ = y u_\tau / \nu$, k turbulent kinetic energy, ε dissipation rate, τ_p particle relaxation time, τ_L fluid Lagrangian integral time, ξ standard normal deviate. Equation numbers in square brackets are those cited in the code comments for Dehbi (2008); they have not been checked against the paper for this revision.

5.1 Particle equation of motion

Fluent integrates the particle momentum equation with the drag law selected in the DPM dialog, computed from the mean slip velocity. That setting is not shown in the reproduced dialogs; Fluent's default "spherical" law is the Morsi–Alexander correlation, whereas the code comment (line 718) states that Schiller–Naumann is the default. The difference is small at the particle Reynolds numbers of this case ($Re_p \lesssim 1$). The UDF adds the drag of the fluid velocity fluctuation as a body force per unit mass, computed with the Schiller–Naumann correction (lines 644–738):

$$\frac{F_i}{m_p} = f_D \, u'_i, \qquad f_D = \frac{1 + 0.15\, Re_p^{0.687}}{\tau_p}, \qquad \tau_p = \frac{\rho_p d_p^2}{18 \mu}, \qquad Re_p = \frac{|\mathbf{U} - \mathbf{u}_p| \, d_p}{\nu}.$$

The fluctuation u'_i is updated once per particle time step, on the call for the first velocity component, and reused for the other two.

5.2 Normalised Langevin equation

The model evolves the normalised fluctuation $\eta_i = u'_i / \sigma_i$:

$$d\eta_i = -\frac{\eta_i}{\tau_L} dt + A_i \, dt + \sqrt{\frac{2}{\tau_L}} \, dW_i .$$

It is discretised semi-implicitly: the relaxation term is implicit, and the drift and noise terms are explicit (lines 397–399, 427–429):

$$\eta_i^{n+1} = \frac{\eta_i^n + \sqrt{2\,\Delta t/\tau_L}\,\xi_i + A_i\,\Delta t}{1 + \Delta t/\tau_L}.$$

The code comments call this “implicit Euler”; it is a semi-implicit (linearly implicit) Euler–Maruyama scheme. For $\Delta t \ll \tau_L$, which holds here because $\Delta t \le 10^{-6}$ s, the stationary variance of η is $1/(1 + \Delta t/2\tau_L) \approx 1$, provided the deviates ξ are independent (see Section 7.2).

5.3 Near-wall branch ($y^+ < 100$)

The near-wall equations are solved in a body-fitted frame (lines 225–275): $\mathbf{e}_2$ is the inward wall normal, $\mathbf{e}_1$ the local mean-velocity direction made orthogonal to $\mathbf{e}_2$, and $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$. The rms fluctuations are algebraic fits to channel-flow DNS data given by Dehbi (2008) [Eqs. 37–39] (lines 131–148):

$$\frac{\sigma_1}{u_\tau} = \frac{0.40\,y^+}{1 + 0.0239\,(y^+)^{1.496}}, \quad \frac{\sigma_2}{u_\tau} = \frac{0.0116\,(y^+)^2}{1 + 0.203\,y^+ + 0.00140\,(y^+)^{2.421}}, \quad \frac{\sigma_3}{u_\tau} = \frac{0.19\,y^+}{1 + 0.0361\,(y^+)^{1.322}}.$$

The code comments attribute these fits to Dreeben and Pope (1997); that attribution could not be confirmed and is not repeated here.

The Lagrangian time scale follows Kallio and Reeks (1989) [Eqs. 31–33] (lines 196–214):

$$\tau_L = \tau_L^+ \frac{\nu}{u_\tau^2}, \qquad \tau_L^+ = \begin{cases} 10, & y^+ \le 5, \\ 7.122 + 0.5731\,y^+ - 0.00129\,(y^+)^2, & 5 < y^+ \le 100. \end{cases}$$

Only the wall-normal component carries a drift term [Eq. 22] (lines 421–422). It is reduced for inertial particles by the local Stokes number $St = \tau_p/\tau_L$ (variable `Stk` in the code; not the same as the flow Stokes number Stk of Table 3):

$$A_1 = A_3 = 0, \qquad A_2 = \frac{1}{1 + St}\,\frac{\partial \sigma_2}{\partial y} = \frac{1}{1 + St}\,\frac{u_\tau^2}{\nu}\,\frac{d(\sigma_2/u_\tau)}{dy^+}.$$

The derivative $d(\sigma_2/u_\tau)/dy^+$ is evaluated analytically (lines 163–188); no gradient computed by Fluent is used. The fluctuation is $u'_i = \eta_i \sigma_i$ in the body-fitted frame and is rotated to Cartesian components.

5.4 Bulk branch ($y^+ \ge 100$)

Away from the wall the turbulence is treated as isotropic (lines 444–474), with $\sigma = (2k/3)^{1/2}$ [Eq. 26] and

$$\tau_L = \frac{2}{C_0}\,\frac{k}{\varepsilon}, \qquad C_0 = 14 \quad \text{[Eq. 36]}.$$

The bulk drift term of the model, proportional to ∇k [Eq. 27], is set to zero in this implementation. The code comment gives the reason: reading the cell gradient of k under the RSM crashed Fluent. The effect of omitting the term was not quantified.

5.5 Wall data and time step

The on-demand function `init_wall_udm` (lines 498–625) stores u_τ, y^+ and the wall normal in user-defined memory. It computes the wall shear stress from the tangential velocity of the wall-adjacent cell divided by the wall distance. The particle time step is $\Delta t = \max[\min(10^{-6}\text{ s}, 0.1\,\tau_p), 10^{-9}\text{ s}]$, capped by Fluent’s own step (lines 765–797).

6. Results

6.1 Continuous phase

The curvature of the bend produces the expected secondary flow: the centrifugal pressure gradient drives faster core fluid towards the outer wall and slower near-wall fluid inwards, forming a pair of counter-rotating Dean vortices. Revision 1 describes this flow qualitatively; no secondary-flow contours or velocity profiles are included in either revision.

6.2 Deposition fraction

Table 3 lists the deposition fractions from the Fluent DPM summaries reproduced in Appendix B of revision 1 (also in `data/deposition_fraction.csv`). The flow Stokes number is $Stk = \rho_p d_p^2 U/(9\mu D)$ with the values of Table 1 and ρ_p = 1000 kg m^{-3}, neglecting the Cunningham correction.

Table 3. Deposition fraction (DF) per particle diameter.

d_p (µm)	Stk	Tracked	Escaped	Deposited (“aborted”)	DF (%)	Sampling standard error (%)
10	0.092	10,000	8,329	1,671	16.7	0.37
20	0.368	10,000	4,279	5,721	57.2	0.49
30	0.829	10,000	2,142	7,858	78.6	0.41
40	1.473	10,000	546	9,454	94.5	0.23
50	2.302	10,000	199	9,801	98.0	0.14

The standard error, $\sqrt{DF(1-DF)/N}$, covers only the finite number of particles. It does not include the effects of the issues in Section 7 or of the settings listed in Section 4.

Revision 1 (Figure 6) also shows points at 5 µm (about 5 %) and 15 µm (about 33 %). No DPM summary is available for these two diameters, so they are not included in Table 3. Revision 1 lists six diameters in its Sec. 4.3, plots seven, and refers to “seven distinct particle diameters (10 to 50 µm)” in its conclusions.

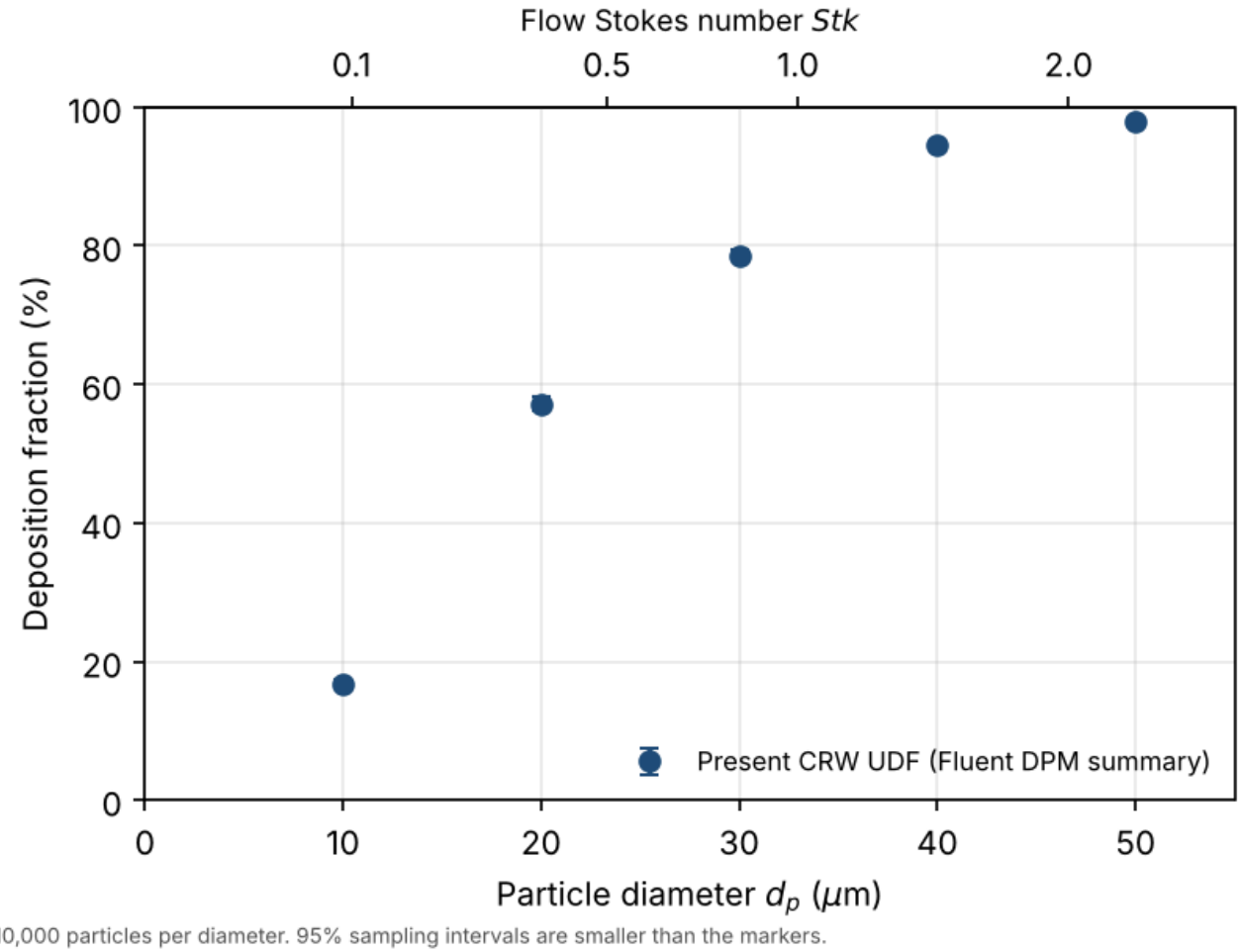


Figure 5. Deposition fraction from Table 3 against particle diameter (bottom axis) and flow Stokes number (top axis), generated by `tools/plot_deposition.py`. The 95 % sampling intervals are smaller than the markers.

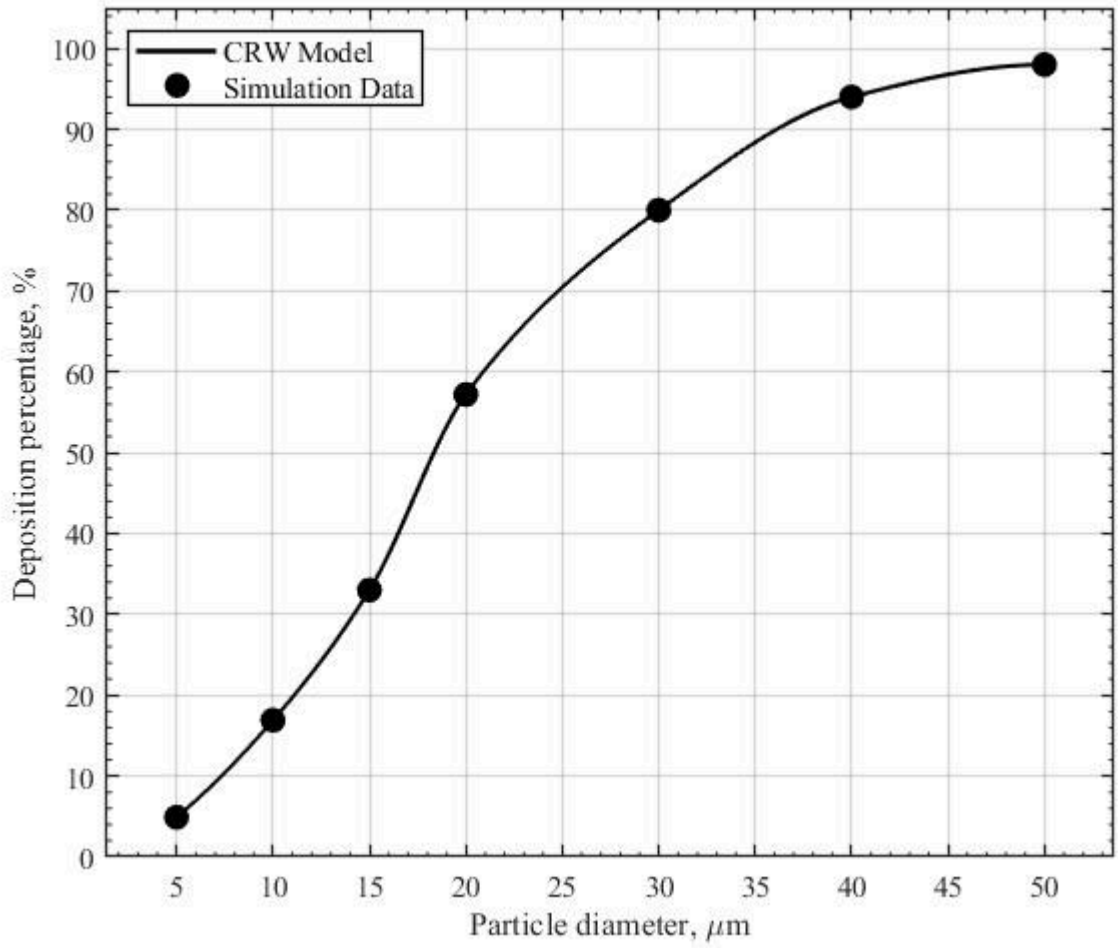


Figure 6. Figure 10 of revision 1, reproduced unchanged.

6.3 Comparison with the benchmark

In Figure 6 the simulated points are drawn with a solid curve labelled “CRW Model”, which revision 1 describes as the numerical curve of Dehbi (2008). The curve passes through every simulated point, including changes of slope at 15 and 20 µm, and no experimental points are shown. The figure therefore does not let the reader judge the agreement, and no agreement is claimed in this revision. The comparison still to be made is:

1. Digitise the Dehbi (2008) results and the Pui et al. (1987) data, with the digitisation uncertainty.
2. Plot them against Table 3 as a function of Stk, because the flow conditions differ (Section 2).
3. Report the difference at each Stokes number.

7. Code review of the UDF

The source file was reviewed line by line for this revision. The file in this repository is unchanged from the version used for the results above. Revision 1 (Appendix A) refers to the source file as `lanudf.c`; the UDF names hooked in Figure S3 match `crw_dehbi2008_rev10.c`.

7.1 The near-wall branch is applied only in wall-adjacent cells

`init_wall_udm` loops over wall faces and writes wall data only into the cell adjacent to each face (`F_C0`, lines 522–618). Every other cell keeps the flag `UDM_INIT = 0`. In `advance_langevin`, such cells fall back to the bulk branch: lines 376–383 compute a y^+ from the wall distance and then overwrite it with `BL_YPLUS_MAX + 1`.

The anisotropic statistics and the drift correction of Section 5.3 therefore act only in the first cell layer, where y^+ is of order 1. All other cells, including those between $y^+ \approx 1$ and $y^+ = 100$ (Section 2: most of the cross-section), use the isotropic, drift-free bulk branch, which is the treatment the Dehbi model was designed to replace. Any agreement with the benchmark obtained with this version could not be attributed to the drift correction as described in Section 5.3.

A second limitation applies even where the branch is active: y^+, σ_i, τ_L and A_2 are read from the cell (`C_UDMI(c, …)`, lines 367–368, 412 and 421), so they are constant within a cell. Inside the first cell, where deposition is decided, σ_2 is frozen at its cell-centre value instead of falling to zero at the wall. Extending the cell data to all cells with $y^+ \le 100$ would still give a piecewise-constant σ_2 with jumps between cells, and the analytical drift would then not match the profile seen by the particle.

Correction for revision 3: store, for every cell with $y^+ \le 100$, the centroid and normal of its nearest wall face and u_τ there, and compute the particle's own wall distance $y_p = (\mathbf{x}_p - \mathbf{x}_w) \cdot \mathbf{n}_w$ at each step. Verify the result with the standard well-mixed test: tracer particles ($\tau_p \to 0$) released uniformly in a straight pipe must remain uniformly distributed.

7.2 The random-number generator produces correlated deviates

`gauss_rand` (lines 94–121) keeps no generator state. It hashes the particle's memory address, the current particle time and a salt through one step of two linear congruential generators and applies the Box–Muller transform. The program `tools/rng_check.c` runs an exact copy of the function. Only the lowest 31 bits of each congruential step are used, so the result is the same whether `unsigned long` has 32 bits (MSVC) or 64 bits.

Table 4. Statistics of the UDF generator, time step 10^{-6} s.

Statistic	Ideal value	Result
Mean, variance (one address, 2×10^5 samples)	0, 1	+0.001, 1.003
Autocorrelation, lags 1 to 5 (address 0x7ff6a1b2c3d0)	0 ± 0.005	+0.22, +0.06, −0.11, −0.29, −0.23
Correlation between two particles at the same time	0 ± 0.005	between −0.65 and +0.85, depending on the address difference
Variance of the sum of $M = 1000$ successive deviates divided by M (six addresses)	1	0.26 to 10.3
As above, 2000 pseudo-random 16-byte-aligned addresses	1	median 0.48; 5–95 % range 0.20 to 2.7; extremes 0.08 and 60

The one-point statistics are correct, but successive deviates are strongly correlated. The last row is the relevant measure for dispersion: over a time of the order of τ_L (about 10^3 steps of 10^{-6} s), the accumulated forcing of a typical particle has about half the variance of independent deviates, and across particles the ratio ranges over almost three orders of magnitude (0.08 to 60), depending only on the particle's memory address. The effective turbulent diffusivity of the model is therefore wrong by a factor that differs from particle to particle. It is also not known whether Fluent reuses the same `Tracked_Particle` structure for successive particles in steady tracking; if it does, particles would receive nearly identical forcing sequences.

Correction for revision 3: give each particle its own generator state, for example a 32-bit xorshift or SplitMix state stored in an additional particle user scalar and seeded once at injection, and extend `rng_check` to test the new generator over many seeds before the simulations are re-run.

7.3 The Langevin state changes frame between branches

In the near-wall branch η holds components in the body-fitted frame ($\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$; lines 427–429). In the bulk branch it holds Cartesian components (x, y, z; lines 465–467). The same three particle variables store both (lines 478–480), and no rotation is applied when a particle moves from one branch to the other. Streamwise, wall-normal and spanwise components are then reinterpreted as x, y and z. Because of Section 7.1, particles near the wall cross between the branches often.

Correction for revision 3: store the dimensional Cartesian fluctuation and re-normalise it in the current frame at each step, $\eta = \mathbf{R}^{\mathsf{T}}\mathbf{u}'/\boldsymbol{\sigma}$.

7.4 Smaller points

1. `dehbi_wall_bc` returns `PATH_ABORT` (lines 749–752). Fluent counts these particles as “aborted”, the same category as particles stopped for numerical reasons such as the step limit (Section 4). Fluent’s built-in trap condition, or a counter in user memory, would separate the two.
2. The deposition comment (lines 741–748) states that a particle reaching a wall face is within one radius of the wall. Fluent detects wall contact when the particle centre reaches the face, so the criterion is applied at zero distance, not at one radius.
3. The bulk-branch comment states that omitting the ∇k drift has a “negligible effect on the final results”. This was not tested.
4. Comments that disagree with the code: the header refers to “d(sigma^2)/dy” (line 10) while the code uses $d\sigma_2/dy$; line 494 promises a “k-eps fallback” for cells away from the wall, but lines 378–382 force the bulk branch and the computed y^+ is overwritten; the generator is described as producing “uncorrelated” numbers (lines 85–86).
5. The integration uses `TP_DT(p)` (line 359) while the update test uses the time since the last update (line 691); these can differ under accuracy control. The first step after injection always uses $u' = 0$ (lines 679–683).
6. The code does not check that 6 user-defined memory locations and 8 particle user scalars are allocated; if they are not, Fluent fails at run time.
7. At $y^+ = 100$ the model switches abruptly between branches. With the equilibrium log-law estimates $k^+ = C_\mu^{-1/2} \approx 3.3$ and $\varepsilon^+ = 1/(\kappa y^+)$, σ_2/u_τ changes from 0.98 to about 1.5 and τ_L^+ from 51.5 to about 20, with no drift term for these jumps. The bulk branch also uses the isotropic $(2k/3)^{1/2}$ although the RSM provides the individual normal stresses.
8. Comment FIX-10 (lines 87–90) describes the conversion of `TP_TIME(p)*1e9` to a 32-bit `unsigned long` as an intentional truncation. A floating-point value outside the range of the target type gives undefined behaviour in C. It does not occur here, because the residence time is below 4.29 s.

The code is otherwise well organised: it uses named constants for memory indices, guards against unassigned cells at injection, has no mutable static state, follows the C89 declaration rules required by the Fluent compiler, and records each change (FIX-2 to FIX-19) in comments.

8. Conclusions

1. A CRW–Langevin dispersion model following Dehbi (2008) was implemented as a compiled ANSYS Fluent UDF with anisotropic near-wall statistics, a body-fitted frame, a Stokes-number-weighted wall-normal drift and an adaptive particle time step.
2. On an RSM flow field, the UDF gives deposition fractions that rise from 16.7 % at 10 µm ($Stk = 0.09$) to 98.0 % at 50 µm ($Stk = 2.3$) (Table 3).
3. Deposition of 5 and 10 µm particles was sensitive to the near-wall mesh; the sensitivity was observed but not quantified.
4. The code review shows that, in the version used, the near-wall branch acts only in wall-adjacent cells and with cell-constant statistics, the random deviates are correlated, and the Langevin state is not rotated between frames.

The case also differs from the Pui et al. (1987) experiment. The results should therefore be described as the output of a working implementation, not as a validation. Revision 3 will correct the three code issues, settle the open settings, and re-run all diameters with a benchmark comparison at matched Stokes number.

9. Use of software tools

As stated in revision 1, the Cursor code editor was used during UDF development to debug C syntax, memory-access errors and compilation problems with Fluent's `udf.h` macros, and to tidy comments. The physical model, equations and coupling logic were taken from Dehbi (2008).

References

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